

Discretization of Errors

Unitary Qubit Errors

$$U = \alpha I + \beta X + \gamma Y + \delta Z$$

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$

$$U_k = \alpha I_k + \beta X_k + \gamma Y_k + \delta Z_k \rightarrow \text{Error on } k\text{th Qubit}$$

$$U_k = I \otimes \underbrace{U}_{\substack{\text{at } k\text{th position}}} \otimes I$$

$$Y = iXZ$$

$$U_k |4\rangle = \alpha |4\rangle + \beta X_k |4\rangle + \gamma i X_k Z_k |4\rangle + \delta Z_k |4\rangle$$

Just before measuring the syndrome, the state of the qubit can be thought as

$$\alpha|4\rangle \otimes |I \text{ syndrome}\rangle + \beta x_k |4\rangle \otimes |X \text{ syndrome}\rangle \\ + \gamma i x_k z_k |4\rangle \otimes |X_k Z_k \text{ syndrome}\rangle + \delta z_k |4\rangle \otimes |Z \text{ syndrome}\rangle$$

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→ In general, after the correction, we'll obtain the state

$$|4\rangle \otimes \epsilon$$

where, $\epsilon = |2|^2 |I \text{ syndrome}\rangle + |\beta|^2 |X \text{ syndrome}\rangle + |\gamma|^2 |X Z \text{ syndrome}\rangle + |\delta|^2 |Z \text{ syndrome}\rangle$

Arbitrary Qubit Errors

$$\Phi(\sigma) = \sum_j A_j \sigma A_j^\dagger \quad \text{Any error}$$

$$A_j = \alpha_j I + \beta_j X + \gamma_j Y + \delta_j Z$$

$$\Phi(|\psi\rangle\langle\psi|) = \sum_j (\alpha_j I + \beta_j X + \gamma_j Y + \delta_j Z) |\psi\rangle\langle\psi| (\alpha_j I + \beta_j X + \gamma_j Y + \delta_j Z)^\dagger$$

→ Following the same procedure, after correction, we get the state,
 $|\psi\rangle\langle\psi| \otimes \{$

where,

$$\varepsilon = \sum_j \left(|\alpha_j|^2 |I_{\text{syndrome}}\rangle\langle I_{\text{syndrome}}| + |\beta_j|^2 |X_{\text{syndrome}}\rangle\langle X_{\text{syndrome}}| + |\gamma_j|^2 |Y_{\text{syndrome}}\rangle\langle Y_{\text{syndrome}}| + |\delta_j|^2 |Z_{\text{syndrome}}\rangle\langle Z_{\text{syndrome}}| \right)$$

This procedure generalises to other Q.E.C. codes